

Parameter-Space Trajectory Metrics

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Overview

Gradient descent is a physical process. Parameters move through a high-dimensional space under forces determined by the loss landscape, the architecture, and the update rule. Standard training diagnostics—loss curves, accuracy, gradient norms—are scalar projections of this movement that discard almost everything.

The framework below instruments the *trajectory* itself: how far did each parameter travel, in what direction, driven by which gradient sources, and how much of that travel was purposeful versus wasted? Three leverage metrics then weight those trajectories by different notions of parameter importance.

Two foundational tensions. Gross and net are not redundant, and neither are efficiency and productivity. The gap between gross path and net displacement is where geometric waste lives—motion that cancelled itself. The gap between directedness and loss improvement is where productive waste lives—motion that was internally consistent but extrinsically useless. The framework is organized around these two tensions.

The metrics decompose into two orthogonal qualities of movement:

- **Intrinsic quality** (Sections 1–2): was the motion internally consistent? Did parameters move in coherent directions, or did they oscillate and cancel? These metrics are geometric—they describe the shape of the trajectory in parameter space without reference to the loss landscape.
- **Extrinsic quality** (Section 3): was the motion productive? Did it actually descend the loss landscape? These metrics require loss evaluations. They are stacked on top of the intrinsic metrics as a multiplicative qualification layer: any intrinsic metric M admits a loss-qualified twin $M^* = M \cdot \Delta^*$, where Δ^* is the fractional loss improvement. The geometric metric tells you about the shape of the trajectory; the qualified metric tells you whether that shape was worth it.

This qualification principle applies uniformly. The intrinsic metrics stand on their own—a training run with high η_1 but low Δ^* has real information: the model moved purposefully, it just did not learn much. The qualified metrics add the extrinsic dimension. Both layers are always worth tracking.

0.1 Motivation

- **Parameters are coordinates; functions are the thing.** Reparameterize a network and the function is unchanged, the parameter trajectory is scrambled. The parameter numbers are arbitrary labels on the same underlying object. Function space is where the object lives.
- **The causal arrow runs through function space.** Loss is a functional quantity. It does not care about parameter values directly—only about what function those values implement. Gradients are the translation layer: “this function is wrong” → “move these numbers.” The Jacobian is the Rosetta Stone between levels.
- **Selection operates at the upper level; mechanism lives below.** Gradient descent is functional selection implemented in parameter coordinates. Which parameter configurations survive is determined entirely by their functional consequences. This is the same structure as: mental activity selects neural routing (Hofstadter), fold selects codon string (evolution), behavior selects synaptic weight (Hebb). The lower level is the mechanism; the upper level is where the reasons live.
- **The DNA / codon analogy.** Hamming distance between codons is a real fact about the string. It is causally inert. What matters is whether the fold changed, whether the binding site moved, whether the protein still recognizes the receptor. Parameter distance is the Hamming distance. Functional displacement is the fold change. The Jacobian is the folding rules.
- **The Jacobian as Rosetta Stone / vielbein.** It is not merely a sensitivity measure. It is the local coordinate transformation between levels—the object that says “a step of this size here means a change of that size there.” In GR this is the vielbein (translates curved spacetime $\leftrightarrow$ local inertial frame). Here it translates parameter nudges $\leftrightarrow$ function-space arrows. The Fisher metric is what you get when you make this into an inner product: $g_{ij} = \langle \partial_i f, \partial_j f \rangle$.
- **The strange loop (Hofstadter hook).** The upper level (function / loss) reaches back down and selects which lower-level configurations (parameters) persist. The lower level (parameter updates) implements the upper-level reasons. Neither is primary. The loop is the system. Gradient descent is a strange loop: functional error becomes numerical update becomes functional correction.
- **Other instances of the same level-translation beauty.**
 - Compile-time tensors → electricity
 - Hunger → muscle stimulation
 - Bad semantics in an LLM → weight update
 - Good traits → favored genetics
 - Mental activity → neural routing

In each case: the upper level is causally legible; the lower level is mechanistically complete. The translation between them is always a Jacobian-like object (sensitivity, gain, fitness gradient, synaptic plasticity rule).

- **Why this framework is philosophically non-trivial.** Most training diagnostics live in parameter space because that is where the optimizer operates and the numbers are cheap to log. This is like studying evolution by counting nucleotide substitutions without ever asking whether the protein changed. The present framework insists on measuring in the space where causality is legible. The entire technical apparatus— $\varepsilon_i(t)$, η_R , leverage redistribution at grokking, the Full-Fisher Lift— follows from this single insistence.
- **The thesis, stated bluntly.** Gradient descent is functional selection. Parameter space is the implementation substrate. Function space is the right level of description. The Jacobian connects them. This framework measures learning in function space, using the Jacobian as the instrument.

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1 Base Accumulation: Gross Path and Net Displacement

Let $\theta_i(t)$ denote the value of parameter element i at training step t , and let $\Delta\theta_i(t) = \theta_i(t) - \theta_i(t-1)$ be the update applied at step t .

Gross path length.

$$G_i = \sum_t |\Delta\theta_i(t)| \quad (1)$$

The total distance parameter i has traveled through parameter space, counting every step in its absolute value. This is the *odometer reading*—it accumulates regardless of direction changes. A parameter that walks +0.1, then −0.1, then +0.1, then −0.1 has $G_i = 0.4$.

Net displacement.

$$D_i = \sum_t \Delta\theta_i(t) = \theta_i(T) - \theta_i(0) \quad (2)$$

The signed sum of all updates: the straight-line vector from the initial position to the current position. The wandering parameter above has $D_i = 0$.

The G – D gap as foundational diagnostic. $G_i \geq |D_i|$, with equality only when the parameter walks a monotone straight line. Every higher-level metric in this document inherits this design: gross quantities measure work done (including wasted, oscillatory, or fought-over work); net quantities measure work committed to the trajectory; their ratios surface different kinds of inefficiency.

Network-level efficiency: two distinct ratios. Two questions are worth separating:

1. *Waste vs. purposefulness*: across all parameters, what fraction of total motion was directed rather than cancelled?
2. *Geodesic-ness*: how close was the network’s overall trajectory to a Euclidean straight line from $\theta(0)$ to $\theta(T)$?

These are answered by different norms:

$$\eta_1 = \frac{|D|_1}{G} = \frac{\sum_i |D_i|}{\sum_i G_i} \in [0, 1] \quad (3)$$

$$\eta_2 = \frac{|D|_2}{G} \in [0, 1] \quad (4)$$

where $G = \sum_i G_i$ is the total gross path. η_1 is the *waste metric*: it equals 1 when every parameter’s motion is perfectly directed, and approaches 0 when updates cancel within each parameter. η_2 is the *geodesic metric*: it additionally penalizes dispersion of directed motion across parameters, shrinking whenever D_i point in different directions even if each $|D_i|/G_i$ is large.

The relationship $\eta_2 \leq \eta_1 \leq 1$ holds always. The gap $\eta_1 - \eta_2$ measures dispersion of directed motion across parameters—structural information about how coordinated learning is, but not waste.

A key connection: η_1 is exactly the G_i -weighted mean of per-parameter efficiencies:

$$\eta_1 = \sum_i \frac{G_i}{G} \cdot \eta_i \quad (5)$$

η_2 does *not* admit this decomposition. Readers should not assume averaging η_i recovers either network-level ratio without checking which norm is in use.

Per-parameter efficiency.

$$\eta_i = \frac{|D_i|}{G_i} \in [0, 1] \quad (6)$$

$\eta_i \rightarrow 1$: the parameter walked a near-straight line. $\eta_i \rightarrow 0$: it churned without going anywhere. The distribution of $\{\eta_i\}$ across the network—and how it shifts by layer, parameter type, and gradient source—is a fine-grained diagnostic not captured by the network-level scalars.

1.1 Per-Source Attribution

When training uses multiple gradient sources (e.g., a trunk loss $\mathcal{L}_0$ and auxiliary head losses $\mathcal{L}_1, \dots, \mathcal{L}_H$), the total gradient at step t is

$$g_i(t) = \sum_{s=0}^{S-1} g_i^{(s)}(t),$$

where $S = H + 1$ is the total number of sources. A single Adam step is applied using the combined gradient $g_i(t)$, preserving unified moment estimates. The actual parameter update is then attributed proportionally:

$$\Delta\theta_i^{(s)}(t) = \frac{g_i^{(s)}(t)}{g_i(t)} \cdot \Delta\theta_i(t) \quad (7)$$

Approximation caveat. This proportional split is exact in the sense that pre-Adam source proportions are preserved post-Adam. However, Adam’s moment estimates $\hat{m}_i$ and $\hat{v}_i$ are accumulated on the *combined* gradient history. Per-step attribution is exact, but the integrated $D_i^{(s)}$ is path-dependent: a late-arriving source inherits the adaptive scaling geometry carved by sources that dominated earlier in training. The attribution should be read as “which source drove this step,” not “what would this source have done alone.”

Per-source gross path and net displacement:

$$G_i^{(s)} = \sum_t |\Delta\theta_i^{(s)}(t)| \quad (8)$$

$$D_i^{(s)} = \sum_t \Delta\theta_i^{(s)}(t) \quad (9)$$

By construction $\sum_s D_i^{(s)} = D_i$ exactly.

1.2 Per-Source Per-Parameter Efficiency

$$\eta_i^{(s)} = \frac{|D_i^{(s)}|}{G_i^{(s)}} \in [0, 1] \quad (10)$$

When $G_i^{(s)} = 0$, set $\eta_i^{(s)} = \text{NaN}$ and exclude from aggregations. This is the primary observable for the recruitment hypothesis: a parameter where the task gradient churns (low $\eta_i^{(0)}$) while the structured head gradient is directed (high $\eta_i^{(s)}$, $s > 0$) is one where the auxiliary signal is doing geodesic work the task signal cannot.

2 Steepness: The Extrinsic Dimension of Movement

2.1 The Core Observation

The efficiency metrics of Section 1 are *intrinsic*: they describe the geometry of the parameter trajectory without any reference to the loss landscape. A perfectly directed trajectory ($\eta_1 = 1$) earns a perfect intrinsic score whether it walks straight into a deep minimum or straight along a flat plateau.

The missing dimension is *extrinsic productivity*: how steeply did the loss descend per unit of parameter motion? A displacement vector has both direction and magnitude. The intrinsic metrics capture direction (was it coherent?). Steepness captures productive magnitude (did it go somewhere useful?).

Consider two training runs reaching the same final loss:

- Run A: $\eta_1 = 0.9$, loss improvement = 0.05. Highly directed, shallow descent.
- Run B: $\eta_1 = 0.4$, loss improvement = 0.80. Chaotic, but steep descent.

Run A is not obviously preferable. Efficient shallow descent may indicate the model is navigating a plateau precisely but not productively. Run B may be stumbling toward a deep minimum. The framework’s job is to make these trade-offs visible, not to resolve them.

2.2 Two Steepnesses

Steepness asks how much loss improvement a unit of movement purchased—but there are two natural denominators, corresponding to gross and net motion respectively, and they answer different questions.

Gross steepness σ_G : loss per unit of effort.

$$\sigma_G = \frac{\Delta\mathcal{L}}{G} = \frac{\mathcal{L}(0) - \mathcal{L}(T)}{G} \quad (11)$$

Loss improvement per unit of total gross parameter motion. This qualifies the odometer: of all the steps taken, including the wasted ones, how much did training descend? High σ_G means each unit of parameter work, even the oscillatory work, bought real loss improvement. This is the right denominator when asking about the productivity of *effort*.

Net steepness σ_D : loss per unit of committed displacement.

$$\sigma_D = \frac{\Delta\mathcal{L}}{|D|_1} \quad (12)$$

Loss improvement per unit of net displacement. This qualifies the straight-line movement: of the distance the model actually committed to traveling, how steeply did it descend? High σ_D means the committed direction was well-aimed. This is the right denominator when asking about the productivity of *direction*.

The relationship between them. Since $\eta_1 = |D|_1/G$, we have exactly:

$$\sigma_D = \frac{\sigma_G}{\eta_1} \quad (13)$$

A highly directed run ($\eta_1 \approx 1$) has $\sigma_D \approx \sigma_G$: most effort was committed, so effort-productivity and direction-productivity are nearly the same. A chaotic run ($\eta_1 \ll 1$) has $\sigma_D \gg \sigma_G$: the committed direction was actually steep, even though most effort was wasted on oscillation. The three quantities η_1 , σ_G , and σ_D are not independent—any two determine the third—but each is the natural answer to a distinct question.

Per-step versions. At training step t , using batch loss as a proxy for test loss (fast but conflates train/test):

$$\sigma_G(t) = \frac{\mathcal{L}(t-1) - \mathcal{L}(t)}{\sum_i |\Delta\theta_i(t)|} \quad \sigma_D(t) = \frac{\mathcal{L}(t-1) - \mathcal{L}(t)}{\sum_i |\Delta\theta_i^{\text{net}}(t)|} \quad (14)$$

Both can be negative on a bad step. Accumulated over training, $\sum_t \sigma_G(t) \cdot G(t) = \Delta\mathcal{L}_{\text{total}}$, recovering total loss improvement as a weighted sum of per-step gross steepnesses.

2.3 The Steepness–Directedness Decomposition

Together, σ_G (gross steepness) and η_1 (directedness) decompose the quality of a training run along two independent axes:

	High σ_G (steep)	Low σ_G (shallow)
High η_1 (directed)	Ideal: purposeful steep descent	Plateau navigation: efficient but unproductive
Low η_1 (chaotic)	Productive thrashing: noisy but learning	Wasted motion: neither directed nor improving

The “plateau navigation” region is particularly insidious: high η_1 can mask stagnation. The “productive thrashing” region (small-batch early training) is where most productive learning occurs in practice—the framework makes this visible rather than penalizing it as waste.

3 The Qualification Principle

3.1 Loss as an Extrinsic Quality of Motion

The intrinsic metrics (G , D , η_1 , η_2 , η_i) describe the geometry of the parameter trajectory. They are defined entirely in terms of parameter updates—no loss evaluations required. Their signal is real: a training run with high η_1 moved purposefully, regardless of whether that purposeful motion improved the model.

But purposeful motion and productive motion are different things. The loss landscape is the extrinsic judge: it does not care how internally consistent the parameter updates were; it only cares where the parameters ended up. The gap between directed motion and productive motion is exactly the gap between memorization and generalization—a network can move very purposefully toward a memorized solution that generalizes poorly.

The qualification principle: any intrinsic metric M admits a loss-qualified twin $M^\dagger = M \cdot \Delta^*$, where:

$$\Delta^* = \max\left(1 - \frac{\mathcal{L}_{\text{test}}(T)}{\mathcal{L}_{\text{test}}(0)}, 0\right) \in [0, 1] \quad (15)$$

is the fractional improvement in held-out test loss from initialization to end of training. $\Delta^* = 0$ means the model learned nothing (or degraded); $\Delta^* = 1$ means the model drove test loss to zero. The $\max(\cdot, 0)$ floor prevents negative values when test loss increases.

Why test loss. Using training loss for Δ^* rewards overfitting. A model can drive training loss to zero by memorizing; its Δ_{train}^* would be high while Δ_{test}^* reveals the problem. The framework is interested in learning quality.

Implementation note. Measure $\mathcal{L}_{\text{test}}(0)$ on the freshly initialized model before the optimizer takes any steps. This baseline is architecture-and-data-dependent but hyperparameter-independent—every configuration in a grid sweep shares the same baseline.

3.2 Qualified Efficiency

The primary qualified metric is *qualified efficiency*:

$$\boxed{Q = \eta_1 \cdot \Delta^*} \quad (16)$$

$Q \in [0, 1]$. A model that moved with perfect directedness but learned nothing scores $Q = 0$. A model that learned maximally but moved with maximum churn also scores $Q = 0$. The jointly optimal model—directed motion that produces steep loss improvement—scores near $Q = 1$.

The relationship to steepness is:

$$Q = \eta_1 \cdot \Delta^* \approx \eta_1 \cdot \frac{\sigma_G \cdot G}{\mathcal{L}(0)} \quad (17)$$

So Q is approximately the product of directedness and gross steepness, normalized by initial loss scale. The triple (η_1, σ_G, Q) is coherent: directedness, productivity-per-unit-effort, and their joint outcome.

3.3 The Qualification Principle Applied Broadly

The same Δ^* factor can be stacked onto any intrinsic metric to produce its qualified twin. A few canonical examples:

Qualified gross path.

$$G^\dagger = G \cdot \Delta^* \quad (18)$$

Total parameter work, discounted by how much of it produced real learning. High $G^\dagger$ requires both significant parameter movement and significant test-loss improvement.

Qualified net displacement.

$$D^\dagger = |D|_1 \cdot \Delta^* \quad (19)$$

Committed parameter movement, discounted by productivity. This is related to Q by $D^\dagger = Q \cdot G$: qualified efficiency times total gross path recovers qualified net displacement.

Qualified per-source efficiency.

$$Q^{(s)} = \eta_1^{(s)} \cdot \Delta^{*(s)} \quad (20)$$

where $\Delta^{*(s)}$ is the fractional loss improvement on the steps where source s dominated. This answers: when this gradient source was steering, how directed was the motion, and how productive were those steps?

Qualified leverage-weighted attribution.

$$\mathcal{B}^{(s)\dagger} = \mathcal{B}^{(s)} \cdot \Delta^* \quad (21)$$

Leverage-weighted gross motion by source, discounted by overall learning productivity. This qualifies ϕ and ψ (Section 7) in the same way.

The general rule. Any metric in this framework that describes how parameters moved can be multiplied by Δ^* to ask whether that movement was worth it. The intrinsic and qualified versions are both worth tracking; their ratio is Δ^* itself, which measures how much of the geometric behavior translated into actual learning.

3.4 TOPSIS Distance: A Complementary Ranking

For comparing configurations across a hyperparameter grid, a second scalar is useful: the *TOPSIS distance* to the ideal point ($\mathcal{L} = 0$, $\eta_1 = 1$). After normalizing both axes to $[0, 1]$ across the comparison set:

$$d_{\text{TOPSIS}} = \sqrt{\ell_{\text{norm}}^2 + (1 - \eta_{1,\text{norm}})^2} \quad (22)$$

Lower distance is better. Q and d_{TOPSIS} are complementary: Q is an absolute product (interpretable without a comparison set), while d_{TOPSIS} is a relative ranking (requires a set of configurations to normalize over). Both should be tracked.

Code note. For a grid sweep over batch size $\times$ learning rate: compute Q for each cell (absolute), then compute d_{TOPSIS} across the full grid for ranking. Plot both as heatmaps. The efficiency ridge—the locus of configurations with high Q —should appear as a diagonal band in the batch-size $\times$ LR space, consistent with the linear scaling rule relating optimal LR to batch size.

4 Metric I: Positional Leverage

$$\lambda_i^{\text{pos}} = \frac{|\{j : j \text{ is downstream of } i\}|}{P} \in [0, 1] \quad (23)$$

Pure graph topology: the fraction of parameters that depend on θ_i through the computation graph. Value-independent and loss-independent. A weight in an early layer feeds into many downstream computations and has high λ^{pos} ; a weight in the final layer has low λ^{pos} .

Vanishing gradient counterpoint. This metric and gradient magnitude point in opposite directions for early layers. Early parameters have maximum positional leverage but minimum gradient magnitude. Positional leverage measures *potential* influence regardless of whether the current values allow that influence to propagate.

$$\tilde{G}_i^{\text{I}} = \lambda_i^{\text{pos}} \cdot G_i \quad (24)$$

5 Metric II: Instantaneous Functional Influence

$$\lambda_i^{\text{fn}}(t) = \left\| \frac{\partial f}{\partial \theta_i}(t) \right\|_2^2 = \sum_{j=1}^p \left(\frac{\partial f_j}{\partial \theta_i}(t) \right)^2 \quad (25)$$

where f is the functional output of the network: $f(x) = \log \text{softmax}(z(x)) \in \mathbb{R}^C$ for classification, or $f(x) \in \mathbb{R}^D$ for regression, and $p = C$ or $p = D$ respectively. This is the diagonal of the Fisher Information Matrix evaluated at the current parameter values $\theta(t)$ —the total squared sensitivity of the model’s output to θ_i right now, summed symmetrically over all output dimensions without reference to any label, loss, or argmax.

Value-dependent and live: unlike λ_i^{str} , which averages out initialization-dependence to expose architectural structure, $\lambda_i^{\text{fn}}(t)$ is a snapshot. It reflects not just where θ_i sits in the graph but what the current weight configuration allows it to do. The same parameter position can have very different λ_i^{fn} at different points in training.

Relationship to the training gradient.

$$\frac{\partial \mathcal{L}}{\partial \theta_i} = \frac{\partial \mathcal{L}}{\partial f} \cdot \frac{\partial f}{\partial \theta_i}$$

Functional influence is the “structural half” of the loss gradient—the part depending on the parameter’s role in the function, separated from the part depending on the loss geometry. Vanishing gradients arise from either a flat loss surface ($\partial \mathcal{L} / \partial f$ small) or a small Jacobian ($\partial f / \partial \theta_i$ small). Metric II isolates the second factor, making it diagnosable independently of the loss.

Riemannian interpretation. The Fisher Information Matrix $\mathcal{F} = \mathbb{E}[J^\top J]$, where $J = \partial f / \partial \theta$, is the natural Riemannian metric tensor on parameter space. Moving in direction v costs $\sqrt{v^\top \mathcal{F} v}$ under this metric. $\lambda_i^{\text{fn}}(t)$ is the i -th diagonal entry of $\mathcal{F}$ at the current parameter values—a diagonal approximation to the natural metric, computed without ever materializing the full matrix.

Computation.

$$\lambda_i^{\text{fn}}(t) \approx \frac{1}{|X_{\text{probe}}|} \sum_{x \in X_{\text{probe}}} \sum_{j=1}^p \left(\frac{\partial f_j(x)}{\partial \theta_i}(t) \right)^2 \quad (26)$$

Evaluated at checkpoints, not inside the training loop. The same probe set X_{probe} used for λ_i^{str} should be reused here for consistency: both metrics are then expectations over the same input distribution, so $\rho_i(t) = \lambda_i^{\text{fn}}(t)/\lambda_i^{\text{str}}$ is a clean ratio without confounding from different x draws. This requires p backward passes per checkpoint per probe example—run at coarse intervals, not every step.

Data dependence. Because $\lambda_i^{\text{fn}}(t)$ is not averaged over initializations, it inherits variance from both the current weight values and the probe inputs. A single representative example is riskier here than for λ_i^{str} : the current network may have learned input-specific structure that makes a single x a poor summary. The probe set average is more important here than in the structural case.

$$\tilde{G}_i^{\text{II}}(t) = \lambda_i^{\text{fn}}(t) \cdot G_i(t) \quad (27)$$

6 Metric III: Structural Potential

$$\lambda_i^{\text{str}} = \mathbb{E}_{\substack{\theta \sim P_{\text{init}} \\ x \sim P_{\text{data}}}} \left[\left\| \frac{\partial f}{\partial \theta_i} \right\|_2^2 \right] \quad (28)$$

where f is the functional output of the network: $f(x) = \log \text{softmax}(z(x)) \in \mathbb{R}^C$ for classification, or $f(x) \in \mathbb{R}^D$ for regression. The squared norm then expands as:

$$\left\| \frac{\partial f}{\partial \theta_i} \right\|_2^2 = \sum_{j=1}^p \left(\frac{\partial f_j}{\partial \theta_i} \right)^2 \quad (29)$$

where $p = C$ (output classes) for classification and $p = D$ (output dimensions) for regression. This is the diagonal of the Fisher Information Matrix evaluated at θ —or equivalently, the total squared sensitivity of the model’s output distribution to θ_i , summed symmetrically over all output dimensions. No labels, no loss, no argmax: f_j is simply the j -th component of the network’s functional output.

The expected Jacobian norm of parameter θ_i under the initialization distribution and the data distribution. You cannot do calculus without values—but you can average over values. As you average over many random initializations and input examples, idiosyncratic value- and example-dependent fluctuations cancel and the structural signal persists: the mean propagation gain from position i to the output, determined by layer type, activation function, depth, and connectivity.

Computation.

$$\lambda_i^{\text{str}} \approx \frac{1}{K} \sum_{k=1}^K \frac{1}{|X_{\text{probe}}|} \sum_{x \in X_{\text{probe}}} \left\| \frac{\partial f^{(k)}(x)}{\partial \theta_i} \right\|_2^2 \quad (30)$$

Computed *once* before training begins; reused as a fixed scaling factor throughout. The expectation is approximated jointly over K random initializations and a fixed probe set X_{probe} drawn once from the data distribution. A probe set of 64–256 examples is the practical sweet spot: large enough to suppress input-dependent variance, small enough that K full Jacobian evaluations remain tractable. Using the full dataset is correct but rarely necessary; using a single mean feature vector is fast but can miss activation-pattern variance across examples.

Heavy precomputation (large K) is warranted. Report standard error $\sigma_i/\sqrt{K}$ alongside the mean; flag parameters where median and mean diverge substantially (heavy-tailed distributions near saturating nonlinearities). For flagged parameters, use a trimmed mean as the denominator of $\rho_i(t)$.

Network-level leverage budget. Define the total structural leverage of the network:

$$\Lambda = \sum_i \lambda_i^{\text{str}} \quad (31)$$

and the per-parameter leverage share:

$$\pi_i = \frac{\lambda_i^{\text{str}}}{\Lambda} \in [0, 1], \quad \sum_i \pi_i = 1 \quad (32)$$

π_i is the fraction of the network’s total functional leverage residing in parameter θ_i . It defines a probability distribution over parameters—the architecture’s intrinsic prior on which parameters matter most. Layer-level aggregates $\pi^{(\ell)} = \sum_{i \in \ell} \pi_i$ give a leverage budget per layer: what share of the network’s total functional surface area does each layer control?

Connection to mean field theory. Poole et al. (2016) derived the per-layer analog analytically, showing exponential decay with depth in the “ordered phase” and growth in the “chaotic phase,” with a critical “edge of chaos” boundary. Metric III is the empirical per-parameter version of the same quantity, applicable to arbitrary architectures without analytical tractability assumptions.

$$\tilde{G}_i^{\text{III}} = \lambda_i^{\text{str}} \cdot G_i \quad (33)$$

6.1 Realized-to-Potential Ratio

$$\rho_i(t) = \frac{\lambda_i^{\text{fn}}(t)}{\lambda_i^{\text{str}}} \quad (34)$$

$\rho_i \approx 1$: the parameter is being used roughly as the architecture intends. $\rho_i \ll 1$: suppression—high structural potential not currently realized (vanishing gradients, saturated activations, dead units). $\rho_i \gg 1$: amplification—the parameter is in a “hot” configuration, exerting more functional influence than its architectural average predicts.

Because λ_i^{str} is fixed and $\lambda_i^{\text{fn}}(t)$ is live, $\rho_i(t)$ cleanly separates what the architecture affords from what training has realized. A parameter with high π_i and low $\rho_i(t)$ is structurally important but currently suppressed—a candidate for targeted intervention. A parameter with low π_i and high $\rho_i(t)$ is being over-recruited relative to its architectural role.

Watching $\rho_i(t)$ evolve over training, decomposed by gradient source, reveals which training objectives recruit structurally important parameters and which spend budget on structurally peripheral ones.

7 Movement in Function Space

The metrics of Sections 1–3 are defined in parameter space: they measure how far parameters travel, how coherently, and with what architectural leverage. But parameters are coordinates. Coordinates are arbitrary. A network can be reparameterized—rescaling adjacent layers, permuting neurons, absorbing constants into biases—to produce an identical function with a completely different parameter trajectory. Any metric defined purely on $\Delta\theta$ is coordinate-dependent in exactly this sense: it measures something real about the optimizer’s behavior in a particular coordinate system, but nothing intrinsic to the function being learned.

The Jacobian $\partial f/\partial\theta_i$ is the object that restores invariance. It is the *pushforward*—the local linear map from parameter space into function space. Two different parameterizations of the same function have different $\Delta\theta$ but the same $J \cdot \Delta\theta$, because the Jacobians transform inversely to the coordinates. Function-space displacement is the coordinate-independent object. It is also the natural one: gradients originate in function space, pass backward through the Jacobian to move parameters, and the Jacobian is therefore the right instrument to project that movement back into function space to assess what it actually accomplished. The accounting closes on itself in a way that raw parameter distance cannot.

Formally, the Jacobian equips parameter space with a Riemannian metric:

$$g_{ij}(\theta) = \left\langle \frac{\partial f}{\partial\theta_i}, \frac{\partial f}{\partial\theta_j} \right\rangle = \mathbb{E}_x! \left[\sum_k \frac{\partial f_k}{\partial\theta_i} \frac{\partial f_k}{\partial\theta_j} \right] \quad (35)$$

This is the Fisher Information Matrix. Moving in direction v costs $\sqrt{v^\top G v}$ in function space under this metric. $\lambda_i^{\text{fn}}(t)$ is the i -th diagonal entry of G at time t —the local coordinate leverage of parameter θ_i : how much a unit step in the θ_i direction moves the network’s output. The derivative is not merely a sensitivity measure; it *is* the local coordinate transformation, the object that defines what a step in parameter space means for the function.

In the continuous limit, gradient descent is a geodesic flow in this Riemannian geometry. Finite step sizes introduce discretization error that breaks the geometric interpretation. The function-space trajectory metrics below are a discrete Euler approximation to the continuous path integral of function-space velocity—exact at infinitesimal learning rate, and increasingly accurate as step size decreases.

7.1 Function-Space Gross Path and Net Displacement

At each training step t , parameter θ_i moves by $\Delta\theta_i(t)$ and the network’s Jacobian is $\lambda_i^{\text{fn}}(t)$. The instantaneous function-space velocity of parameter θ_i is $\lambda_i^{\text{fn}}(t) \cdot |\Delta\theta_i(t)|$ —the product of local leverage and parameter displacement. Integrating over training:

$$G_i^f = \sum_t \lambda_i^{\text{fn}}(t) \cdot |\Delta\theta_i(t)| \quad (36)$$

G_i^f is the *function-space gross path* of parameter θ_i : the total function-space distance its updates have traversed, counting every step regardless of direction. This is the discrete Euler approximation to the path integral:

$$G_i^f \approx \int_0^T \left| \frac{\partial f}{\partial \theta_i}(t) \right|_2 \cdot |\dot{\theta}_i(t)|, dt \quad (37)$$

The only approximation is the checkpoint frequency of λ_i^{fn} : computed at every step, this is exact; computed at coarse intervals, it assumes the Jacobian is piecewise constant between checkpoints. Near phase transitions—where λ_i^{fn} can spike rapidly—coarse checkpointing may miss fast Jacobian changes. This is a feature, not a bug, as a diagnostic: large discrepancy between coarse and fine estimates of G_i^f is itself a signal that a phase transition occurred in that interval.

The signed analog accumulates direction:

$$D_i^f = \sum_t \lambda_i^{\text{fn}}(t) \cdot \Delta \theta_i(t) \quad (38)$$

D_i^f is the *function-space net displacement* of parameter θ_i : positive when the parameter pushed the function consistently in one direction in output space, cancelling when it reversed. The ratio:

$$\eta_i^f = \frac{|D_i^f|}{G_i^f} \in [0, 1] \quad (39)$$

is the *function-space efficiency* of parameter θ_i —strictly stronger than η_i . Where η_i asks whether the parameter moved coherently in weight space, η_i^f asks whether that movement produced coherent displacement in function space. A parameter can have high η_i (directionally consistent updates) but low η_i^f if those consistent updates happened to push the output in alternating directions—for instance, if the Jacobian reversed sign between steps. η_i^f is the right measure of whether a parameter’s training history was functionally purposeful.

7.2 Network-Level Function-Space Trajectory

Summing over all parameters gives the network-level function-space odometer:

$$G^f = \sum_i G_i^f \quad (40)$$

and the network-level function-space efficiency:

$$\eta_1^f = \frac{\sum_i |D_i^f|}{G^f} \in [0, 1] \quad (41)$$

These are the function-space analogs of G and η_1 , and the same two-tension decomposition applies: G^f measures total function-space work done (including wasted and oscillatory work); η_1^f measures what fraction of that work was directionally committed. Their ratio surfaces geometric waste in function space rather than parameter space.

The qualification principle extends directly. The function-space qualified efficiency:

$$Q^f = \eta_1^f \cdot \Delta^* \quad (42)$$

asks whether function-space motion was both coherent and productive. Q^f is a strictly more meaningful scalar than Q : it measures the same joint optimum of directedness and learning, but in the coordinate-invariant units of the output space rather than the arbitrary units of the parameter coordinates.

7.3 Realized-to-Potential Ratio in Function Space

The structural potential λ_i^{str} defines what function-space influence parameter θ_i affords in expectation under the initialization distribution. The leverage-weighted gross path $\tilde{G}_i^{\text{III}} = \lambda_i^{\text{str}} \cdot G_i$ is therefore the function-space gross path the parameter *would have* accumulated if it had always operated at its structural average influence. The ratio:

$$\rho_i^f(T) = \frac{G_i^f}{\lambda_i^{\text{str}} \cdot G_i} = \frac{\tilde{G}_i^{\text{II}}}{\tilde{G}_i^{\text{III}}} \quad (43)$$

is the *function-space realized-to-potential ratio*. $\rho_i^f > 1$: training found a configuration where this parameter punched above its architectural weight, accumulating more function-space displacement than its structural average predicts. $\rho_i^f < 1$: the parameter was structurally capable of more function-space influence than training realized—suppressed by vanishing gradients, saturated activations, or competition from other parameters. ρ_i^f separates the architectural question (what does this position afford?) from the training question (how much of that did training use?), in the units that actually matter.

7.4 Function-Space Leverage Share

Define the per-parameter share of total function-space gross path:

$$\pi_i^f(t) = \frac{G_i^f(t)}{G^f(t)} \quad (44)$$

$\pi_i^f(t)$ is the fraction of the network’s total function-space displacement that parameter θ_i is responsible for, accumulated up to time t . Unlike the structural share π_i (which is fixed before training begins), π_i^f is dynamic: it evolves as training reshapes the Jacobian. Watching $\pi_i^f(t)$ evolve reveals which parameters are recruited into functional relevance over the course of learning, and when.

Layer-level aggregates $\pi^{f,(\ell)}(t) = \sum_{i \in \ell} \pi_i^f(t)$ give a dynamic function-space budget per layer: what share of the network’s total functional surface area did each layer control at each phase of training?

7.5 Phase Transitions as Jacobian Redistribution

The function-space trajectory metrics make a specific prediction about phase transitions in learning—most concretely, grokking power2022grokking.

Before generalization, a network that has memorized a training set implements something close to a lookup table. Functional responsibility is spread diffusely across parameters: many have moderate λ_i^{fn} because no clean circuit has formed. G^f is accumulating, but it

is incoherent—high G_i^f , low D_i^f , low η_i^f across the memorizing parameter population. The function-space leverage share π_i^f is approximately uniform across parameters.

When the generalizing circuit crystallizes, the Jacobian concentrates. A small subset of parameters—those implementing the discovered regularity—suddenly acquire large λ_i^{fn} because those directions in parameter space now connect directly to the output through a clean computational path. The memorizing parameters’ λ_i^{fn} simultaneously decreases as they are bypassed by the new circuit. The result is a rapid *redistribution* of π_i^f : total function-space leverage reorganizing across the network. This is not a perturbation in parameter velocity—weight decay may be quietly shrinking memorization weights while the generalizing weights have already found their position, leaving parameter-space velocity smooth or even decreasing. The phase transition is invisible in G_i and η_i but present as a spike and redistribution in $\lambda_i^{\text{fn}}(t)$ and $\pi_i^f(t)$.

This gives a concrete, falsifiable cross-validation against mechanistic interpretability results. nanda2023progress identified the Fourier features underlying modular arithmetic generalization and the epoch at which they crystallize. The function-space trajectory framework predicts:

1. $\lambda_i^{\text{fn}}(t)$ spikes for the Fourier-feature parameter subset at the generalization epoch, with a simultaneous drop for the memorizing parameters.
2. $\pi_i^f(t)$ undergoes a redistribution—a rewiring of the function-space metric—coinciding with the accuracy phase transition.
3. The learning order map Ξ (Section 8) shows a cohort of training examples whose per-example efficiency $\xi_n(t)$ spikes at the same epoch, corresponding to the examples whose structure is captured by the crystallizing Fourier circuit.

If these three signatures align with the activation-space interpretability results, then trajectory-space and activation-space interpretability are confirmed to be observing the same event from complementary angles: one watching the geometry of the parameter space rewire, the other watching the representations that geometry produces.

8 The Full-Fisher Lift

The function-space metrics of the preceding sections use $\lambda_i^{\text{fn}}(t)$ —the i -th diagonal entry of the Fisher Information Matrix $G(t) = \mathbb{E}_x[J^\top J]$ —as the measure of each parameter’s functional leverage. This is a deliberate approximation, and it is worth being precise about what it assumes and what it misses.

The first-order change in the network’s output induced by a parameter update $\Delta\theta$ is:

$$\Delta f \approx J \Delta\theta = \sum_i \frac{\partial f}{\partial \theta_i} \Delta\theta_i \quad (45)$$

The squared magnitude of this change, averaged over the data distribution, is:

$$\mathbb{E}[\|\Delta f\|^2] = \Delta\theta^\top G \Delta\theta = \underbrace{\sum_i G_{ii} \Delta\theta_i^2}_{\text{diagonal terms}} + 2 \underbrace{\sum_{i < j} G_{ij} \Delta\theta_i \Delta\theta_j}_{\text{off-diagonal terms}} \quad (46)$$

The diagonal terms G_{ii} measure how much the output moves when θ_i is perturbed alone. The off-diagonal terms G_{ij} measure the co-directionality of the output perturbations caused by θ_i and θ_j jointly: $G_{ij} > 0$ means the two parameters reinforce each other’s functional effect; $G_{ij} < 0$ means they partially cancel. The diagonal approximation—summing per-parameter scalar displacements as in G_i^f and D_i^f —implicitly assumes that all parameters affect the function in mutually orthogonal directions. This is false in general, and emphatically false inside circuits, where the entire computational point is that parameters cooperate. The “magic” of network computation—cooperation, redundancy, and cancellation among parameters—lives in the off-diagonal terms.

The diagonal metrics are nonetheless the right object for the *architectural* question: λ_i^{str} , the structural potential, is computed as an expectation over random initializations, where off-diagonal terms average to zero and the diagonal is the true signal. The diagonal is also cheaper, interpretable at the individual parameter level, and sufficient for many diagnostic purposes. The full-Fisher lift is not a replacement but a completion: it gives the proper coordinate-invariant objects for training dynamics, while the diagonal metrics remain the right lens for architectural analysis.

8.1 The Natural Riemannian Line Element

Define the *instantaneous functional velocity* of a parameter update $\Delta\theta_t$:

$$v(t) = \sqrt{\Delta\theta_t^\top G(t) \Delta\theta_t} = \sqrt{\mathbb{E}_x[\|J(t) \Delta\theta_t\|^2]} \quad (47)$$

$v(t)$ is the natural Riemannian line element under the Fisher metric. It measures the actual magnitude of the output-space perturbation caused by the parameter update at step t , averaged over the data distribution. Crucially, it is *coordinate-invariant*: two different parameterizations of the same function yield the same $v(t)$, because the Jacobians transform inversely to the coordinate change and the product $J \Delta\theta$ is preserved. This is the invariance that the diagonal approximation sacrifices.

The *total functional path length* traversed by the network over training is:

$$L = \sum_t v(t) \quad (48)$$

L is the function-space odometer: the total distance the network’s function has traveled through output space, in units that are intrinsic to the function rather than to any particular parameterization. It replaces the diagonal-only G^f of the preceding section.

In the continuous limit, L is the Riemannian arc length of the training trajectory under the Fisher metric:

$$L \approx \int_0^T \sqrt{\dot{\theta}(t)^\top G(\theta(t)) \dot{\theta}(t)} dt \quad (49)$$

The discrete sum is the Euler approximation to this path integral, exact at infinitesimal learning rate and increasingly accurate as step size decreases.

8.2 Per-Parameter Functional Attribution

To decompose $v(t)$ into additive contributions from individual parameters, observe that the total linearized output change is $\Delta f_t \approx \sum_i c_i(t)$, where:

$$c_i(t) = \left. \frac{\partial f}{\partial \theta_i} \right|_t \Delta \theta_i(t) \quad (50)$$

is the vector in output space that parameter θ_i contributes to the step. The step length is $v(t) = \|\Delta f_t\|$ (expectation over x understood). Projecting each $c_i(t)$ onto the total output change gives a signed scalar attribution:

$$\varepsilon_i(t) = \frac{c_i(t) \cdot \Delta f_t}{v(t)} \quad \text{if } v(t) > 0, \quad 0 \text{ otherwise} \quad (51)$$

By construction $\sum_i \varepsilon_i(t) = v(t)$: the attributions sum exactly to the step’s functional length. $\varepsilon_i(t)$ is signed: positive when parameter θ_i pushes in the net functional direction of the step, negative when it opposes the dominant direction. It is the *only* additive decomposition of the step’s functional length that respects the actual geometry of output space.

This replaces the diagonal approximation $\lambda_i^{\text{fn}}(t) \cdot \Delta \theta_i(t)$, which treats each parameter’s output push as independent. $\varepsilon_i(t)$ accounts for the full interaction structure: a parameter whose output direction aligns with the network-wide functional step is credited; one that fights the consensus is debited.

From $\varepsilon_i(t)$ we define the full-Fisher analogs of the base accumulation metrics:

$$\mathcal{E}_i = \sum_t |\varepsilon_i(t)| \quad (\text{functional gross effort}) \quad (52)$$

$$\mathcal{N}_i = \sum_t \varepsilon_i(t) \quad (\text{functional net contribution}) \quad (53)$$

$$\eta_i^f = \frac{|\mathcal{N}_i|}{\mathcal{E}_i} \in [0, 1] \quad (\text{per-parameter functional efficiency}) \quad (54)$$

$\mathcal{E}_i$ is the total functional effort expended by parameter θ_i over training, counting all pushes regardless of whether they aligned with the network’s consensus direction. $\mathcal{N}_i$ is the net functional contribution: how much of that effort was consistently directed. η_i^f is the ratio—a measure of whether the parameter’s functional history was purposeful, in units that respect parameter interactions.

8.3 Network-Level Full-Fisher Efficiency

Two complementary network-level efficiency metrics follow from the per-parameter attributions.

Inter-parameter coherence.

$$\eta_{\text{net}}^f = \frac{L}{\sum_i \mathcal{E}_i} \in [0, 1] \quad (55)$$

The ratio of total functional path length to total per-parameter functional effort. $\eta_{\text{net}}^f < 1$ whenever parameters fight each other: the individual efforts $|\varepsilon_i(t)|$ sum to more than $v(t)$

because some are positive and some negative, so total effort exceeds net displacement. Low η_{net}^f is a signature of parameter conflict—the network’s components are pulling in different functional directions, wasting effort in cancellation.

Riemannian trajectory efficiency. Let the net functional displacement be:

$$D_{\text{vec}}^f = \sqrt{\mathbb{E}_x[\|f_{\theta(T)}(x) - f_{\theta(0)}(x)\|^2]} \quad (56)$$

the RMS change in the network’s output from initialization to end of training, averaged over the data distribution. Then:

$$\eta_R = \frac{D_{\text{vec}}^f}{L} \in [0, 1] \quad (57)$$

η_R is the *Riemannian trajectory efficiency*: the ratio of straight-line functional displacement to total functional path length. $\eta_R = 1$ iff the network moved along a perfect geodesic in function space—a straight line from its initial function to its final function. $\eta_R \ll 1$ means the function wandered: it traveled far in output space but much of that travel reversed direction.

This is the true function-space analog of η_1 , but immune to coordinate changes and diagonal approximations. Where η_1 asked whether parameter updates were directionally consistent, η_R asks whether the *function itself* moved consistently—a strictly stronger and more meaningful question. The qualification principle applies directly:

$$Q_R = \eta_R \cdot \Delta^* \quad (58)$$

Q_R is the fully coordinate-invariant qualified efficiency: coherent functional motion that produced real learning.

8.4 The Off-Diagonal Structure: Cooperation, Redundancy, and Phase Transitions

The full Fisher matrix $G(t)$ contains information that no per-parameter scalar can capture: the functional relationships *between* parameters. Having access to $G(t)$ —or block-diagonal approximations per layer—opens a set of diagnostics with no analog in the diagonal framework.

Local cooperativity.

$$\chi_{ij}(t) = \frac{G_{ij}(t)}{\sqrt{G_{ii}(t)G_{jj}(t)}} \in [-1, 1] \quad (59)$$

The normalized off-diagonal entry: the cosine similarity between the output-space directions that θ_i and θ_j push. $\chi_{ij} \approx 1$ means the two parameters are functionally redundant—they affect the output in nearly the same direction. $\chi_{ij} \approx -1$ means they are antagonistic. $\chi_{ij} \approx 0$ means they are functionally independent. A sudden clustering of high positive χ_{ij} within a parameter group is a signature of circuit formation: the parameters have aligned their functional effects, indicating they have been recruited into a common computational role.

Layer-wise effective rank. For the Fisher block G_ℓ of layer ℓ :

$$r_{\text{eff}}^{(\ell)}(t) = \frac{(\text{tr } G_\ell)^2}{\text{tr}(G_\ell^2)} \quad (60)$$

The stable rank of the Fisher block: the effective number of independent functional directions the layer is controlling. A sharp drop in $r_{\text{eff}}^{(\ell)}(t)$ signals that functional influence is concentrating onto a low-dimensional subspace of output space—the Fisher block is becoming low-rank as a small group of directions dominates. This is a direct signature of circuit crystallization: the layer’s distributed, diffuse functional influence collapsing into a sparse, specialized computation. Crucially, this signal may appear *before* any movement in the loss or accuracy curves, providing early warning of an impending phase transition.

Redundancy budget.

$$\rho_{\text{off}}^{(\ell)}(t) = \frac{\sum_{i \neq j} |G_{ij}|}{\sum_{i,j} |G_{ij}|} \quad (61)$$

The fraction of the layer’s Fisher mass carried by off-diagonal entries. High $\rho_{\text{off}}^{(\ell)}$ indicates that many parameters are functionally interchangeable—they affect the output in correlated directions and could substitute for each other. A decrease in $\rho_{\text{off}}^{(\ell)}$ marks a transition to specialized roles: parameters are diverging in their functional effects, each carving out a distinct direction in output space. This is the Fisher-block signature of the memorization-to-generalization transition: memorization is functionally redundant (many parameters doing similar things); generalization is functionally specialized (each circuit component doing something distinct).

8.5 Correspondence Table

The full-Fisher lift replaces every diagonal function-space metric with a coordinate-invariant counterpart. The mapping is exact:

Diagonal approximation	Full-Fisher lift
$G_i^f = \sum_t \lambda_i^{\text{fn}}(t) \Delta\theta_i(t) $	$\mathcal{E}_i = \sum_t \varepsilon_i(t) $
$D_i^f = \sum_t \lambda_i^{\text{fn}}(t) \Delta\theta_i(t)$	$\mathcal{N}_i = \sum_t \varepsilon_i(t)$
$\eta_i^f = D_i^f /G_i^f$	$\eta_i^f = \mathcal{N}_i /\mathcal{E}_i$
$G^f = \sum_i G_i^f$	$L = \sum_t v(t)$
η_1^f (diagonal coherence)	η_{net}^f (inter-parameter coherence)
— (no analog)	η_R (Riemannian trajectory efficiency)
$Q^f = \eta_1^f \cdot \Delta^*$	$Q_R = \eta_R \cdot \Delta^*$
$\pi_i^f = G_i^f/G^f$	$\pi_i^f = \mathcal{E}_i/\sum_j \mathcal{E}_j$
— (no analog)	$\chi_{ij}(t)$ (local cooperativity)
— (no analog)	$r_{\text{eff}}^{(\ell)}(t)$ (layer effective rank)
— (no analog)	$\rho_{\text{off}}^{(\ell)}(t)$ (redundancy budget)

The leverage-weighted variants, source attribution, learning order map, and all qualified twins upgrade naturally by substituting $|\varepsilon_i(t)|$ (or its signed value) for $\lambda_i^{\text{fn}}(t) |\Delta\theta_i(t)|$ wherever a per-parameter functional displacement appears. The structural potential λ_i^{str} remains

diagonal by construction: it is an expectation over random initializations, where off-diagonal Fisher entries average to zero and the diagonal is the correct architectural signal.

8.6 Computational Note

The only additional computation required beyond the diagonal metrics is the Jacobian-vector product $J(t) \Delta\theta_t$ at each checkpoint. This is a single backward pass with $\Delta\theta_t$ as a synthetic gradient vector—the same cost as one standard backward pass, not p passes. For architectures where the full Fisher block is needed (cooperativity, effective rank), block-diagonal K-FAC factors computed during training provide a tractable approximation. In research analyses where training dynamics are the primary object of study, the cost is well justified; in production monitoring, the diagonal metrics remain a fast and useful approximation with honest caveats applied.

9 Circuit Discovery via Function-Space Trajectory

The function-space trajectory metrics of the previous section are not merely diagnostic instruments for hyperparameter comparison. They are, in principle, a *causal and temporal record of how a network learned*—and that record contains enough information to reverse-engineer the formation of computational circuits from training dynamics alone, without inspecting activations, without ablations, and without knowing in advance what to look for.

This section develops that claim: why the trajectory record is the right object for circuit discovery, how the discovery procedure works, how it will be tested against a known ground truth, and what confirmation would mean for the framework.

9.1 The Limits of Archaeological Interpretability

Standard mechanistic interpretability is post-hoc. A trained model is frozen; its activations are probed; attention patterns are visualized; activation patching identifies which components are necessary for a behavior. The method is powerful—it has produced genuine circuit-level understanding of induction heads, indirect object identification, and Fourier feature computation in modular arithmetic. But it is fundamentally *archaeological*: it examines the fossil and infers the organism. The trained model is the artifact; the process that produced it is gone.

This limits what post-hoc analysis can answer:

- **When** did a circuit form? Activation patching on a trained model has no temporal axis. It can identify that a circuit exists; it cannot say at what epoch it crystallized, how long formation took, or what the network was doing immediately before.
- **What caused it?** Without knowing when a circuit formed, you cannot study the gradient signal, the data distribution, or the parameter dynamics that immediately preceded formation. Causality requires a timeline.
- **What formed first?** Circuits are not atomic. They have prerequisite structure—sub-circuits that must exist before the full computation can assemble. Post-hoc analysis sees the completed assembly; the order of assembly is invisible.

The function-space trajectory framework is upstream of all of these questions. It produces a time-resolved record of which parameters were doing functional work, when, and how much. The timestamp of circuit formation is not inferred after the fact—it is *observed* as it happens.

9.2 Why Functional Leverage Reveals Circuit Formation

A circuit, in the mechanistic interpretability sense, is a subgraph of the computation graph that implements a specific transformation from inputs to outputs. Before a circuit forms, responsibility for any given output dimension is diffuse—spread across many parameters with moderate λ_i^{fn} , because no clean computational path has been carved. After a circuit forms, a small subset of parameters connects directly and efficiently to the output through the learned structure. Those parameters’ Jacobian norms grow; the bypassed parameters’ Jacobian norms shrink.

This is not a metaphor. It is a direct consequence of the chain rule. If parameter θ_i participates in a circuit that strongly determines output f_k , then $\partial f_k / \partial \theta_i$ is large. If θ_i is in a dead or bypassed branch, $\partial f_k / \partial \theta_i$ is small. Circuit formation is, in the language of this framework, a *redistribution of functional leverage* $\pi_i^f(t)$ —a rewaving of the Jacobian fabric across the parameter population.

Several specific signatures follow:

- **Leverage spike.** At the moment of circuit crystallization, the parameters implementing the circuit show a rapid increase in $\lambda_i^{\text{fn}}(t)$. The rate of change $d\lambda_i^{\text{fn}}/dt$ is large and positive for circuit parameters, large and negative for bypassed memorization parameters.
- **Share redistribution.** $\pi_i^f(t)$ shifts: circuit parameters absorb a growing fraction of total network function-space displacement; peripheral parameters’ shares collapse. The distribution $\pi_i^f(t)$ becomes more concentrated—a measurable decrease in entropy of the leverage distribution.
- **Function-space efficiency spike.** η_i^f rises for circuit parameters at formation: their updates begin producing coherent, directed displacement in output space rather than diffuse oscillation. The parameter was moving before; now its movement *means* something.
- **Temporal alignment with loss.** The leverage redistribution should time-align with the sharpest drop in generalization loss—the moment the circuit begins carrying the computation that determines correct outputs on held-out data.

None of these signatures require knowing which circuit you are looking for. They are blind, bottom-up signals derived purely from the training dynamics.

9.3 Automated Circuit Hypothesis Generation

The above signatures define a concrete, automated procedure for circuit hypothesis generation from training dynamics alone:

1. **Train with frequent Jacobian checkpoints.** Compute $\lambda_i^{\text{fn}}(t)$ at every step or at fine-grained intervals using the probe set X_{probe} . Accumulate $G_i^f(t)$, $D_i^f(t)$, and $\pi_i^f(t)$ continuously. ““
2. **Detect phase transitions.** Identify epochs t^* where the test loss drops sharply—equivalently, where $\Delta^*(t)$ increases steeply. These are the candidate moments of circuit crystallization. A sliding-window slope on test loss is sufficient: flag windows where the slope exceeds a threshold.
3. **Correlate metric slopes with loss slope.** For each flagged window, compute the sliding-window slope of $\lambda_i^{\text{fn}}(t)$ and $\pi_i^f(t)$ for every parameter. Rank parameters by the magnitude of their leverage change during the flagged window:

$$\delta_i^*(t^*) = \frac{1}{W} \sum_{t \in [t^* - W, t^* + W]} \frac{d\lambda_i^{\text{fn}}}{dt} \quad (62)$$

Parameters with large $|\delta_i^*(t^*)|$ are those whose functional role changed most at the phase transition.

4. **Extract candidate circuit parameters.** The top- k parameters by $|\delta_i^*(t^*)|$ are the circuit hypothesis: the parameters the framework predicts were recruited into (positive δ_i^*) or released from (negative δ_i^*) the generalizing computation at this moment.
5. **Confirm with activation-space analysis.** Take the candidate parameter set to standard mechanistic interpretability tools—activation patching, probing, attention visualization—to confirm the hypothesized circuit and characterize the representation it implements. ““

This inverts the standard interpretability workflow. Rather than choosing which components to patch based on prior knowledge or intuition, the trajectory framework generates the candidate list from data. Activation-space analysis becomes *confirmation* rather than *discovery*—a faster and more principled research loop.

9.4 Test Case: Grokking on Modular Arithmetic

The grokking phenomenon power2022grokking provides an ideal testbed: a known phase transition, a known mechanistic explanation, and a known ground truth against which the trajectory framework’s predictions can be checked.

The ground truth. nanda2023progress trained small transformers on modular addition ($a + b \pmod{p}$) and identified the precise mechanism of generalization: a small set of Fourier features, implemented by specific attention heads and MLP neurons, that compute the modular sum via trigonometric identities. They identified the epoch at which these features crystallize and the parameters that implement them.

The predictions. The function-space trajectory framework makes three falsifiable, quantitative predictions:

- **Prediction 1: Leverage spike at the known epoch.** $\lambda_i^{\text{fn}}(t)$ increases sharply for the Fourier-feature parameters identified by Nanda et al. at the generalization epoch they report, with a simultaneous decrease for the memorizing parameters. The spike should be visible as a discontinuity in $d\lambda_i^{\text{fn}}/dt$ that is not present in the parameter velocity $|\dot{\theta}_i|$.
““
- **Prediction 2: Leverage share redistribution.** $\pi_i^f(t)$ concentrates onto the Fourier-feature parameter subset at the generalization epoch. The entropy of the leverage distribution $H[\pi^f(t)] = -\sum_i \pi_i^f(t) \log \pi_i^f(t)$ decreases sharply, reflecting the collapse of diffuse memorization structure into sparse circuit structure.
- **Prediction 3: Learning order map alignment.** The per-example efficiency $\xi_n(t)$ (Section 8) spikes for the cohort of (a, b) pairs whose modular relationships are captured by the crystallizing Fourier features. The epoch of this spike aligns with Predictions 1 and 2 and with the activation-space crystallization epoch of Nanda et al. ““

What confirmation means. If all three predictions are confirmed:

- The framework successfully detected the grokking circuit from training dynamics alone—*without inspecting activations, without prior knowledge of the Fourier mechanism, and without any ablations*. The automated procedure of Section ?? would have nominated the correct parameters as circuit candidates.
““
- Trajectory-space and activation-space interpretability are shown to be observing the same underlying event from complementary angles: one watching the geometry of parameter space rewire, the other watching the representations that geometry produces. Their convergence on the same epoch and the same parameters is a mutual cross-validation.
- The leverage entropy $H[\pi^f(t)]$ is validated as a scalar signature of generalization: a network that has grokked should show markedly lower leverage entropy than one that has merely memorized, reflecting the difference between sparse circuit computation and diffuse lookup.
- The framework offers a new instrument for the grokking phenomenon itself: not just *when* does the phase transition occur (visible in accuracy curves) but *which parameters* drove it, *in what order*, and *at what cost in function-space displacement*. ““

What disconfirmation means. If the leverage spike does not align with the known Fourier-feature parameters or the known epoch, this is equally informative: either the Jacobian checkpointing was too coarse to resolve the transition, the probe set was unrepresentative, or the framework is tracking something real but distinct from the activation-space circuit story. Each failure mode is diagnosable and correctable.

9.5 The General Program

Grokking is the proof of concept. The general program is broader:

- **Any network, any task.** The leverage redistribution signature does not depend on modular arithmetic or Fourier features. Any time a network transitions from a less structured to a more structured computation—any time a circuit crystallizes—the Jacobian fabric should rewire. The procedure of Section ?? is architecture- and task-agnostic. ““
- **Prerequisite ordering.** Because the framework is time-resolved, it can in principle detect the *order* in which sub-circuits form. If circuit B requires circuit A as a prerequisite, A ’s leverage spike should precede B ’s. This is a form of causal ordering that post-hoc analysis cannot recover.
- **Curriculum and data effects.** The learning order map Ξ connects circuit formation to the data: which examples triggered which parameter recruitments. A curriculum that preferentially surfaces the examples whose $\xi_n(t)$ is rising — the gateway examples of Section 8.4—should accelerate circuit formation by providing the gradient signal at the moment the network is primed to use it.
- **Auxiliary supervision as circuit recruitment.** The per-source attribution of Section 1.1 can be combined with the function-space metrics: which gradient source drove the leverage spike? If auxiliary heads grading for intermediate Fourier features produce a λ_i^{fn} spike earlier than the task loss alone, then structured supervision is confirmed to be doing what it is designed to do—recruiting the right parameters into the right circuit before the task signal is strong enough to do it unassisted. This is the function-space evidence for the recruitment hypothesis. ““

The framework is, in this sense, not merely a training diagnostic. It is a proposed instrument for *watching learning happen*—at the parameter level, in the units that matter, with the temporal resolution to ask not just what a network learned but how, when, and why.

10 Leverage-Weighted Source Attribution

Given leverage weights λ_i (from any of Metrics I–III):

$$\mathcal{B}^{(s)} = \sum_i \lambda_i \cdot G_i^{(s)} \quad (63)$$

The total leveraged motion source s has produced. A source that pushed hard at low-leverage parameters has small $\mathcal{B}^{(s)}$; one that pushed at high-leverage parameters has large $\mathcal{B}^{(s)}$.

$$\tilde{D}^{(s)} = \left| \lambda \odot D^{(s)} \right|_2 \quad (64)$$

$$\tilde{\eta}^{(s)} = \frac{\tilde{D}^{(s)}}{\mathcal{B}^{(s)}} \quad (65)$$

Two normalizations of structured-source share:

$$\phi = \frac{\mathcal{B}^{(\text{struct})}}{\sum_s \mathcal{B}^{(s)}} \quad (66)$$

$$\psi = \frac{\mathcal{B}^{(\text{struct})}}{\sum_i \lambda_i \cdot G_i} \quad (67)$$

ϕ normalizes against all demanded leveraged motion (including cancelled work between sources). ψ normalizes against actually committed leveraged motion. $\psi > 1$ is possible and informative: the structured heads pushed for more leveraged motion than the trajectory committed to, meaning their pushes were partially cancelled. The gap $\psi - \phi$ surfaces the degree of inter-source conflict. Track both; they answer different questions.

If ϕ rises as Q rises, the structured heads are the geodesic force.

11 Example-Level Attribution and the Learning Order Map

11.1 Motivation

Everything in Sections 1–7 attributes motion and efficiency to *gradient sources* (which loss function, which head). A complementary and deeper attribution is possible: to *training examples*. Just as backpropagation assigns credit for loss to specific parameters, we can assign credit for learning progress to specific data points.

This is inspired by Prioritized Experience Replay (PER, Schaul et al. 2015), which maintains a per-example priority score and samples high-priority examples more frequently. PER defines priority by current loss magnitude—the highest-loss examples are replayed most. The framework here defines an analogous per-example score grounded in trajectory geometry: *on which examples did the model make steep, well-aimed moves?*

11.2 Per-Example Efficiency Score

After each batch update at step t , the trajectory metrics for that step are available:

- $\eta_1(t)$: directedness of the parameter update
- $\sigma_G(t)$: gross steepness (loss drop per unit gross motion)
- $\sum_i \lambda_i \cdot |\Delta\theta_i(t)|$: leverage-weighted motion magnitude

For each example x_n present in the batch at step t , accumulate a scalar score:

$$\xi_n(t) = \frac{1}{|\mathcal{B}_t|} \cdot Q(t) \cdot \sum_i \lambda_i \cdot |\Delta\theta_i(t)| \quad (68)$$

where $\mathcal{B}_t$ is the batch at step t and the $1/|\mathcal{B}_t|$ factor distributes the step’s credit equally among all examples in the batch. The cumulative per-example score is:

$$\Xi_n = \sum_{t:n \in \mathcal{B}_t} \xi_n(t) \quad (69)$$

This is the total leverage-weighted, loss-qualified, directedness-weighted learning credit attributed to example x_n over training.

Interpretation.

- **High Ξ_n :** steps containing x_n were steep, directed, and leverage-productively weighted. x_n is a *high-yield example*—the model was primed to learn from it when it appeared.
- **Low Ξ_n :** steps containing x_n were chaotic, shallow, or both. x_n is either inherently hard or arrives at the wrong moment in training.
- **Decreasing $\xi_n(t)$ over time:** x_n was productive early but the model has saturated its information content—it taught something specific and is now redundant.
- **Increasing $\xi_n(t)$ over time:** x_n was unlearnable early but becomes highly productive late—its information required prerequisite structure the model had not yet built.

11.3 The Learning Order Map

Track $\xi_n(t)$ for all examples and epochs, producing a matrix $\Xi \in \mathbb{R}^{N \times T}$ where N is the dataset size and T is the number of training epochs. This matrix encodes the *temporal structure of learning*—which examples yielded productive steps, and when.

Analysis operations on Ξ .

1. **Row clustering:** group examples by their temporal efficiency profile. Examples with similar $\xi_n(\cdot)$ curves belong to the same *learning family*—learned together because they share a common structure the model discovered at a particular phase.
2. **Column (epoch) analysis:** the epoch-level mean $\bar{\xi}(t) = \frac{1}{N} \sum_n \xi_n(t)$ tracks how productive training was as a function of time. A spike in $\bar{\xi}(t)$ indicates a phase where many examples suddenly became productive—this is the trajectory-metric signature of grokking.
3. **Correlation with input structure:** in the modular arithmetic setting, examples are fully characterized by (a, b) . Plotting Ξ_n as a function of a and b reveals which arithmetic relationships were learned first and which were learned last.
4. **Prerequisite detection:** if example x_n has a delayed spike in $\xi_n(t)$, examine which examples $\{x_m\}$ had high $\xi_m(t')$ for $t' < t$. These are the candidate *prerequisites*—examples that built the structure x_n required.

11.4 The Four Example Types

The steepness–directedness decomposition applied at the example level produces a taxonomy:

	High $\sigma_{G,n}$ (steep steps)	Low $\sigma_{G,n}$ (shallow steps)
High $\eta_{1,n}$ (directed)	<i>Gateway:</i> model learns cleanly and productively	<i>Plateau:</i> purposeful motion, little loss improvement
Low $\eta_{1,n}$ (chaotic)	<i>Struggle:</i> thrashing but still descending	<i>Wasted:</i> neither directed nor productive

Gateway examples are the highest-yield training data: they appear at the right time, align with what the model is currently capable of learning, and produce steep directed improvement. A curriculum that preferentially samples gateway examples would accelerate learning.

Struggle examples correspond to the PER high-priority class—the model loses a lot on them and updates are noisy. But the trajectory framework distinguishes between struggle examples that are nevertheless descending the loss (productive thrashing) and those that are merely churning. PER cannot make this distinction; the trajectory framework can.

11.5 Connection to Mechanistic Interpretability

The learning order map is mechanistic interpretability from the data side rather than the activation side. Instead of asking “what circuit fired for this example,” it asks “when did this family of examples become efficiently learnable.” These questions are complementary and should converge: a circuit that crystallizes at epoch t_0 should correspond to a cohort of examples whose $\xi_n(t)$ rises sharply at t_0 .

For the modular arithmetic grokking experiment this is directly testable. Nanda et al. (2023) identified the Fourier features and the epoch at which they crystallize. The learning order map should show a cohort of examples whose $\xi_n(t)$ spikes at exactly that epoch. If it does, activation-space interpretability and trajectory-space attribution are confirmed to be seeing the same event from different angles.

12 Trajectory-Guided Hyperparameter Adaptation

12.1 The Observation

Empirically, Q depends on both the hyperparameters and the current state of training. In the early phase, small-batch small-LR configurations produce highest Q : the model is in a chaotic region and low-variance gradient estimates with small step sizes navigate it more productively. In the late phase, larger batches and higher learning rates can sustain high Q because the model has found stable descent directions and can be pushed harder along them.

This suggests that optimal hyperparameters are not static but *trajectory-dependent*: the best batch size and learning rate at epoch t depend on the current $Q(t)$ and its recent history. Standard schedules (cosine decay, cyclical LR) are heuristic approximations to this ideal.

12.2 Finite-Difference Hyperparameter Gradient

Let $Q(h, t)$ denote the qualified efficiency at training step t under hyperparameter configuration $h = (\text{bs}, \text{lr})$. The ideal update is:

$$h_{t+1} = h_t + \alpha_h \cdot \nabla_h Q(h_t, t) \quad (70)$$

The full gradient $\nabla_h Q$ is intractable analytically. A tractable approximation uses finite differences over short horizons:

1. At step t , run N_{probe} steps under h_t ; measure $\bar{Q}_0 = \frac{1}{N_{\text{probe}}} \sum Q(t+k)$.
2. Perturb: run N_{probe} steps under $h_t + \epsilon \hat{e}_{\text{lr}}$; measure $\bar{Q}_{\text{lr}}$.

3. Repeat for $h_t + \epsilon \hat{e}_{\text{bs}}$; measure $\bar{Q}_{\text{bs}}$.
4. Estimate $\nabla_h Q \approx [(\bar{Q}_{\text{lr}} - \bar{Q}_0)/\epsilon, (\bar{Q}_{\text{bs}} - \bar{Q}_0)/\epsilon]$.
5. Update h by a small step in the direction of increasing Q .

This is an online hyperparameter optimization algorithm guided by trajectory metrics rather than validation loss. The key difference: we optimize for *the quality of the learning process*, not just its outcome.

Practical constraint. Probe runs can be run on a small pilot batch rather than the full dataset, trading estimate variance for compute efficiency.

12.3 Q as an Adaptive Stopping Signal

Rather than stopping when validation loss plateaus, stop when Q has been consistently low for M consecutive epochs:

$$\text{stop if } \frac{1}{M} \sum_{k=0}^{M-1} Q(T-k) < Q_{\min} \quad (71)$$

This stops training when neither directedness nor productivity is being achieved. It is more principled than loss plateau detection because it requires *both* conditions to fail simultaneously.

13 Summary Table

Metric	Val.-dep.	Loss-dep.	What it measures
<i>Layers 1–3: Intrinsic (no loss evaluations required)</i>			
G_i, D_i	Yes	No	Odometer and displacement per parameter
$\eta_i = D_i /G_i$	Yes	No	Per-parameter directedness
$\eta_1 = D _1/G$	Yes	No	Network-wide waste (G_i -wtd. mean of η_i)
$\eta_2 = D _2/G$	Yes	No	Network-wide geodesic-ness
$\eta_i^{(s)}$	Yes	No	Per-param per-source directedness
$G_i^{(s)}, D_i^{(s)}$	Yes	No	Gross path and displacement per source
$\mathcal{B}^{(s)}, \phi, \psi$	Yes	No	Leveraged motion share per source
λ_i^{pos}	No	No	Structural potential by graph topology
$\lambda_i^{\text{fn}}(t)$	Yes	No	Realized behavioral influence now
λ_i^{str}	No (avg.)	No	Architectural influence in expectation
$\rho_i(t)$	Yes	No	Realized vs. architectural potential
<i>Layer 4: Extrinsic qualification (loss-dependent)</i>			
Δ^*	Yes	Yes	Fractional test-loss improvement
$\sigma_G = \Delta\mathcal{L}/G$	Yes	Yes	Gross steepness: loss per unit effort
$\sigma_D = \Delta\mathcal{L}/ D _1$	Yes	Yes	Net steepness: loss per unit direction
$Q = \eta_1 \cdot \Delta^*$	Yes	Yes	Qualified efficiency
$Q^{(s)} = \eta_1^{(s)} \cdot \Delta^{*(s)}$	Yes	Yes	Per-source qualified efficiency
$M^\dagger = M \cdot \Delta^*$	Yes	Yes	Any intrinsic metric, loss-qualified
d_{TOPSIS}	Yes	Yes	Distance to ideal ($\mathcal{L} = 0, \eta_1 = 1$)
<i>Layer 5: Example attribution</i>			
ξ_n	Yes	Yes	Per-example step-level learning credit
$\Xi \in \mathbb{R}^{N \times T}$	Yes	Yes	Learning order map

The two tensions, restated. All diagnostic value in this framework lives in one of two gaps:

1. G vs. $|D|$: the geometric gap. How much of the motion was wasted in oscillation? Captured by $\eta_1, \eta_2, \eta_i, \eta_i^{(s)}$.
2. η_1 vs. Q (equivalently, 1 vs. Δ^*): the productive gap. How much of the directed motion translated into actual learning? Captured by σ_G, σ_D, Q , and all qualified twins $M^\dagger$.

The geometric gap is intrinsic—measurable without any loss evaluations. The productive gap is extrinsic—requires held-out loss. Both are necessary; neither alone is sufficient.